

QR decomposition

$$A = QR$$

↳ It is a $m \times n$ matrix

Q is an orthonormal set of vectors which we can find by using Gram-Schmidt process.

$$Q = (q_1 \quad q_2 \quad \dots \quad q_n)$$

* Suppose we have a coefficient matrix

$$A = \begin{bmatrix} 3 & 1 \\ 6 & 2 \\ 0 & 2 \end{bmatrix}$$

$\downarrow \qquad \downarrow$
 $u_1 \qquad u_2$

Let each column as $u_1, u_2, \dots, u_n$

to find Q we need to know two formulas:

$$q_1 = \frac{u_1}{\|u_1\|}$$

$$1. P_k = U_k - \sum_{i=1}^{k-1} (U_k^T q_i) q_i$$

$$2. q_k = \frac{P_k}{|P_k|}$$

↓
Gram-schmidt process

Using these formulas (Gram-schmidt) process we will convert U into q .

One by One step to solve the problem

Step 1 : $k = 1$

$$P_1 = U_1 = \begin{pmatrix} 3 \\ 6 \\ 0 \end{pmatrix}$$

We always start $i \geq 1$ and the summation loop starts from 1 to $k-1$.

Here k is 1 so $k-1=0$

This not possible because we can't go from 1 to 0.

So $\sum_{i=1}^{k-1} (U_k^T q_i) q_i$ won't be added here for P_1

$$q_1 = \frac{P_1}{|P_1|} = \frac{\begin{pmatrix} 3 \\ 6 \\ 0 \end{pmatrix}}{\sqrt{3^2 + 6^2 + 0^2}} = \frac{1}{\sqrt{45}} \begin{pmatrix} 3 \\ 6 \\ 0 \end{pmatrix}$$

Step 2: $k = 2$

$$p_2 = u_2 - \sum_{i=1}^{k-1} (u_2^T q_i) q_i$$
$$= u_2 - \sum_{i=1}^1 (u_2^T q_i) q_i$$

$$= u_2 - (u_2^T q_1) q_1$$

$$= \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} - \left\{ \begin{pmatrix} 1 & 2 & 2 \end{pmatrix} \frac{1}{\sqrt{45}} \begin{pmatrix} 3 \\ 6 \\ 0 \end{pmatrix} \right\} q_1$$

$$= \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} - \frac{15}{\sqrt{45}} \cdot q_1$$

$$= \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} - \frac{15}{\sqrt{45}} \cdot \frac{1}{\sqrt{45}} \begin{pmatrix} 3 \\ 6 \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} - \frac{15}{45} \begin{pmatrix} 3 \\ 6 \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} - \begin{pmatrix} \frac{3}{3} \\ \frac{6}{3} \\ \frac{0}{3} \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 2 \end{pmatrix}$$

$$\begin{aligned}
 q_2 &= \frac{p_2}{|p_2|} \\
 &= \frac{\begin{pmatrix} 0 \\ 0 \\ 2 \end{pmatrix}}{\sqrt{0^2 + 0^2 + 2^2}} \\
 &= \frac{1}{2} \begin{pmatrix} 0 \\ 0 \\ 2 \end{pmatrix} \\
 &= \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}
 \end{aligned}$$

$$Q = \begin{pmatrix} q_1 & q_2 \end{pmatrix}$$

$$= \begin{pmatrix} 3/\sqrt{45} & 0 \\ 6/\sqrt{45} & 0 \\ 0 & 1 \end{pmatrix}$$

$$Q^T = \begin{pmatrix} 3/\sqrt{45} & 6/\sqrt{45} & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$Ax = b$$

(m x n)

Previously we have used Transpose matrix to convert A into a square matrix. Again we will use it to generate the formula for QR decomposition.

$$A^T A x = A^T b$$

$$(QR)^T (QR) x = (QR)^T b$$

$$Q^T R^T Q R x = Q^T R^T b$$

$$\cancel{R}^T (Q^T Q) R x = Q^T \cancel{R}^T b$$

$$\boxed{Q^T Q = I}$$

According to the property of normality

$$\therefore \boxed{R x = Q^T b}$$

This is the final formula. We will use this to solve the system by QR decomposition.

We know,

$$\boxed{A = QR}$$

From the first page

Proper Example 1 :

$$f(-3) = 0$$

$$f(0) = 0$$

$$f(6) = 2$$

Here degree will be $n=2$ but in order to make this an overdetermined system we will consider P_1 , which means we will take $n=1$.

$$P_1 = a_0 + a_1(x) = f(x)$$

$$P_1(-3) = a_0 + a_1(-3) = 0$$

$$P_1(0) = a_0 + a_1(0) = 0$$

$$P_1(6) = a_0 + a_1(6) = 2$$

$$\begin{bmatrix} 1 & -3 \\ 1 & 0 \\ 1 & 6 \end{bmatrix} \begin{matrix} \xrightarrow{A} \\ \\ \end{matrix} \begin{matrix} \begin{bmatrix} -3 \\ 0 \\ 6 \end{bmatrix} \\ \\ \end{matrix} \begin{matrix} \xrightarrow{x} \\ \\ \end{matrix} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{matrix} \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix} \\ \\ \end{matrix} \begin{matrix} \xrightarrow{b} \\ \\ \end{matrix}$$

$U_1 \quad U_2$

In matrix A or coefficient matrix, there are 2 columns now we will have two steps.

Step 1 $\boxed{K = 1}$

$$P_1 = U_1$$

$$= \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

$$q_1 = \frac{P_1}{|P_1|}$$

$$= \frac{\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}}{\sqrt{1^2 + 1^2 + 1^2}}$$

$$= \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

Step 2 $\boxed{K = 2}$

$$P_2 = U_2 - (U_2^T q_1) q_1$$

$$= \begin{pmatrix} -3 \\ 0 \\ 6 \end{pmatrix} - \left\{ (-3 \ 0 \ 6) \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right\} q_1$$

$$= \begin{pmatrix} -3 \\ 0 \\ 6 \end{pmatrix} - \sqrt{3} \cdot q_1$$

$$= \begin{pmatrix} -3 \\ 0 \\ 6 \end{pmatrix} - \sqrt{3} \cdot \begin{pmatrix} 1/\sqrt{3} \\ 1/\sqrt{3} \\ 1/\sqrt{3} \end{pmatrix}$$

$$= \begin{pmatrix} -3 \\ 0 \\ 6 \end{pmatrix} - \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -4 \\ -1 \\ 5 \end{pmatrix}$$

$$q_2 = \begin{pmatrix} \frac{p_2}{|p_2|} \end{pmatrix} = \frac{\begin{pmatrix} -4 \\ -1 \\ 5 \end{pmatrix}}{\sqrt{(-4)^2 + (-1)^2 + 5^2}} = \frac{1}{\sqrt{42}} \begin{pmatrix} -4 \\ -1 \\ 5 \end{pmatrix}$$

$$Q = \begin{pmatrix} q_1 & q_2 \end{pmatrix}$$

$$= \begin{pmatrix} \frac{1}{\sqrt{3}} & \frac{-4}{\sqrt{42}} \\ \frac{1}{\sqrt{3}} & \frac{-1}{\sqrt{42}} \\ \frac{1}{\sqrt{3}} & \frac{5}{\sqrt{42}} \end{pmatrix}$$

$$Q^T = \begin{pmatrix} 1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} \\ -4/\sqrt{42} & -1/\sqrt{42} & 5/\sqrt{42} \end{pmatrix}$$

Now we have to find R

$$R = \begin{bmatrix} U_1^T q_1 & U_2^T q_1 \\ 0 & U_2^T q_2 \end{bmatrix} \rightarrow \text{This is not correct if } U_1 \text{ and } U_2 \text{ is given}$$

$$U_1^T q_1 = \begin{pmatrix} 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1/\sqrt{3} \\ 1/\sqrt{3} \\ 1/\sqrt{3} \end{pmatrix} = \sqrt{3}$$

$$U_2^T q_1 = \begin{pmatrix} -3 & 0 & 6 \end{pmatrix} \begin{pmatrix} 1/\sqrt{3} \\ 1/\sqrt{3} \\ 1/\sqrt{3} \end{pmatrix} = \sqrt{3}$$

$$U_2^T q_2 = \begin{pmatrix} -3 & 0 & 6 \end{pmatrix} \begin{pmatrix} -4/\sqrt{42} \\ -1/\sqrt{42} \\ 5/\sqrt{42} \end{pmatrix} = \sqrt{42}$$

$$\therefore R = \begin{bmatrix} \sqrt{3} & \sqrt{3} \\ 0 & \sqrt{42} \end{bmatrix}$$

$$R\alpha = Q^T b$$

$$\begin{bmatrix} \sqrt{3} & \sqrt{3} \\ 0 & \sqrt{42} \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} \\ -4/\sqrt{42} & -1/\sqrt{42} & 5/\sqrt{42} \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}$$

$$\begin{bmatrix} \sqrt{3} & \sqrt{3} \\ 0 & \sqrt{42} \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} 2/\sqrt{3} \\ 10/\sqrt{42} \end{bmatrix}$$

$$\sqrt{42} a_1 = 10/\sqrt{42}$$

$$a_1 = \frac{10}{\sqrt{42} \times \sqrt{42}}$$

$$= \frac{10}{42}$$

$$\boxed{a_1 = \frac{5}{21}}$$

$$\sqrt{3} a_0 + \sqrt{3} a_1 = \frac{2}{\sqrt{3}}$$

$$\sqrt{3} a_0 + \sqrt{3} \times \frac{5}{21} = \frac{2}{\sqrt{3}}$$

$$\boxed{a_0 = 3/7}$$

Example 2

$$a_0 + a_1 + a_2 = 2$$

$$a_0 + 2a_1 + 4a_2 = 3$$

$$a_0 + 3a_1 + 9a_2 = 6$$

$$a_0 + 4a_1 + 16a_2 = 4$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \\ 1 & 4 & 16 \end{bmatrix} \begin{matrix} A \\ x \end{matrix} = \begin{matrix} b \\ \begin{bmatrix} 2 \\ 3 \\ 6 \\ 4 \end{bmatrix} \end{matrix}$$

→ 4×3 matrix (It is an overdetermined system)

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \\ 1 & 4 & 16 \\ u_1 & u_2 & u_3 \end{bmatrix}$$

There will be 3 steps.

Step 1

$$\boxed{k = 1}$$

$$p_1 = u_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$$

$$q_1 = \frac{p_1}{|p_1|} = \frac{\begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}}{\sqrt{1^2 + 1^2 + 1^2 + 1^2}} = \frac{1}{2} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$$

Step 2

$$\boxed{k = 2}$$

$$p_2 = u_2 - (u_2^T q_1) q_1$$

$$= \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix} - \left\{ (1 \ 2 \ 3 \ 4) \frac{1}{2} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} \right\} \frac{1}{2} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$$

$$= \begin{pmatrix} -1.5 \\ -0.5 \\ 0.5 \\ 1.5 \end{pmatrix}$$

$$q_2 = \frac{p_2}{|p_2|} = \frac{1}{\sqrt{5}} \begin{pmatrix} -1.5 \\ -0.5 \\ 0.5 \\ 1.5 \end{pmatrix}$$

Step 3 :

$$\boxed{K=3}$$

$$P_3 = U_3 - \left[(U_3^T q_1) q_1 + (U_3^T q_2) q_2 \right]$$

$$= \begin{pmatrix} 1 \\ 4 \\ 9 \\ 16 \end{pmatrix} - \left[15 \begin{pmatrix} 0.5 \\ 0.5 \\ 0.5 \\ 0.5 \end{pmatrix} + 5\sqrt{5} \begin{pmatrix} -1.5/\sqrt{5} \\ -0.5/\sqrt{5} \\ -0.5/\sqrt{5} \\ 1.5/\sqrt{5} \end{pmatrix} \right]$$

$$= \begin{pmatrix} 1 \\ -1 \\ -1 \\ 1 \end{pmatrix}$$

$$q_3 = \frac{P_3}{|P_3|} = \frac{1}{2} \begin{pmatrix} 1 \\ -1 \\ -1 \\ 1 \end{pmatrix}$$

$$Q = \begin{pmatrix} 0.5 & -1.5/\sqrt{5} & 0.5 \\ 0.5 & -0.5/\sqrt{5} & -0.5 \\ 0.5 & 0.5/\sqrt{5} & -0.5 \\ 0.5 & 1.5/\sqrt{5} & 0.5 \\ q_1 & q_2 & q_3 \end{pmatrix}$$

$$Q^T = \begin{pmatrix} 0.5 & 0.5 & 0.5 & 0.5 \\ -1.5/\sqrt{5} & -0.5/\sqrt{5} & -0.5/\sqrt{5} & 1.5/\sqrt{5} \\ 0.5 & -0.5 & -0.5 & 0.5 \end{pmatrix}$$

Now For R matrix

$$R = \begin{bmatrix} U_1^T q_1 & U_2^T q_1 & U_3^T q_1 \\ 0 & U_2^T q_2 & U_3^T q_2 \\ 0 & 0 & U_3^T q_3 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 5 & 15 \\ 0 & \sqrt{5} & 5\sqrt{5} \\ 0 & 0 & 2 \end{bmatrix}$$

$$R x = Q^T b$$

$$\begin{bmatrix} 2 & 5 & 15 \\ 0 & \sqrt{5} & 5\sqrt{5} \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 0.5 & 0.5 & 0.5 & 0.5 \\ -1.5/\sqrt{5} & 1.5/\sqrt{5} & 0.5/\sqrt{5} & 1.5/\sqrt{5} \\ -0.5 & -0.5 & -0.5 & 0.5 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \\ 6 \\ 4 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 5 & 15 \\ 0 & \sqrt{5} & 5\sqrt{5} \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 5.5 \\ 27\sqrt{5}/10 \\ -3/2 \end{bmatrix}$$

$$a_2 = -3/4$$

$$\sqrt{5} a_1 + 5\sqrt{5} \times -3/4 = 27\sqrt{5}/10$$

$$a_1 = \frac{129}{20}$$

$$= 6.45$$

$$2a_0 + 5 \times 6.45 + 15 \times (-3/4) = 5.5$$

$$a_0 = -7.75$$

* Note: kindly check the values (calculations)